

DBMS SELLER AND INVENTORY MANAGEMENT

AJIO E-Commerce database management system

1. AJIO SELLER TABLE

1.1 SELLER TABLE

ATTRIBUTE	DESCRIPTION
Seller_ID	Unique seller ID
Seller_Name	Seller name
Email	Seller email address
Phone_Number	Seller contact number
Address	Seller address
GST_Number	Seller GST number

2. AJIO INVENTORY TABLE

ATTRIBUTE	DESCRIPTION
Inventory_ID	Unique inventory ID
Product_ID	Product ID
Seller_ID	Seller ID
Stock_Quantity	Available stock quantity
Stock_Status	Availability status of the product
Last_Updated	Date and time when inventory was last updated

3. Relationship Between Seller and Inventory Tables

- **Seller_ID** is the **Primary Key (PK)** in the **SELLER** table.
- **Seller_ID** is a **Foreign Key (FK)** in the **INVENTORY** table.
- One **Seller** can have **multiple Inventory records**.
- Each **Inventory record** belongs to **one Seller**.
- Therefore, the relationship is **One-to-Many (1:N)**.

4 TABLE CREATION

4.1 SELLER TABLE

```
SQL> CREATE TABLE AJIOSeller (  
2     Seller_ID NUMBER PRIMARY KEY,  
3     Seller_Name VARCHAR2(30),  
4     Email VARCHAR2(50),  
5     Phone_Number VARCHAR2(15),  
6     Address VARCHAR2(80),  
7     GST_Number VARCHAR2(20)  
8 );
```

Table created.

4.2 INVENTORTY TABLE

```
SQL> CREATE TABLE AJIOInventory (  
2     Inventory_ID NUMBER PRIMARY KEY,  
3     Product_ID NUMBER REFERENCES AJIOProduct(Product_ID),  
4     Seller_ID NUMBER REFERENCES AJIOSeller(Seller_ID),  
5     Stock_Quantity NUMBER NOT NULL,  
6     Stock_Status VARCHAR2(20) NOT NULL,  
7     Last_Updated DATE NOT NULL  
8 );
```

Table created.

5. DATA INSERTION

5.1 Seller data insertion

```
SQL> INSERT INTO AJIOSeller VALUES  
2 (301, 'FASHION HUB', 'fashionhub@gmail.com', '9876501234', '12, Anna Nagar, Chennai', '33ABCDE1234F1Z5');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO AJIOSeller VALUES  
2 (302, 'STYLE MART', 'stylemart@gmail.com', '9876501235', '45, Gandhi Road, Tambaram', '33BCDEF2345G2Z6');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO AJIOSeller VALUES  
2 (303, 'TRENDY WEAR', 'trendywear@gmail.com', '9876501236', '78, MG Road, Coimbatore', '33CDEFG3456H3Z7');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO AJIOSeller VALUES  
2 (304, 'URBAN FASHION', 'urbanfashion@gmail.com', '9876501237', '23, Cross Street, Velachery', '33DEFGH4567J4Z8');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO AJIOSeller VALUES  
2 (305, 'FASHION POINT', 'fashionpoint@gmail.com', '9876501238', '56, Bharathi Nagar, Madurai', '33EFGHI5678K5Z9');  
  
1 row created.
```

5.2 Inventory data insertion

```
SQL> INSERT INTO AJIOInventory VALUES
  2 (401, 101, 301, 50, 'Available', TO_DATE('18-09-2026','DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO AJIOInventory VALUES
  2 (402, 102, 302, 35, 'Available', TO_DATE('18-09-2026','DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO AJIOInventory VALUES
  2 (403, 103, 303, 0, 'Unavailable', TO_DATE('18-09-2026','DD-MM-YYYY'));

1 row created.
```

6. Seller Product Information

```
SQL> SELECT
  2     s.Seller_ID,
  3     s.Seller_Name,
  4     p.Product_ID,
  5     p.Product_Name,
  6     p.Price,
  7     i.Stock_Quantity,
  8     i.Stock_Status
  9 FROM AJIOSeller s
 10 JOIN AJIOInventory i
 11 ON s.Seller_ID = i.Seller_ID
 12 JOIN AJIOProduct p
 13 ON i.Product_ID = p.Product_ID
 14 ORDER BY s.Seller_ID;
```

SELLER_ID	SELLER_NAME	PRODUCT_ID
301	FASHION HUB	101
1099	Men Casual Shirt	50 Available
302	STYLE MART	102
799	Women Kurti	35 Available
303	TRENDY WEAR	103
1499	Running Shoes	0 Unavailable

7. Available Products

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     s.Seller_Name,
  5     i.Stock_Quantity,
  6     i.Stock_Status
  7 FROM AJIOInventory i
  8 JOIN AJIOProduct p
  9 ON i.Product_ID = p.Product_ID
 10 JOIN AJIOSeller s
 11 ON i.Seller_ID = s.Seller_ID
 12 WHERE i.Stock_Status = 'Available';
```

PRODUCT_ID	PRODUCT_NAME	SELLER_NAME	STOCK_QUANTITY	STOCK_STATUS
101	Men Casual Shirt	FASHION HUB	50	Available
102	Women Kurti	STYLE MART	35	Available

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
SELLER_NAME          STOCK_QUANTITY STOCK_STATUS
-----
101
Men Casual Shirt
FASHION HUB          50 Available

102
Women Kurti
STYLE MART           35 Available

PRODUCT_ID
-----
PRODUCT_NAME
-----
SELLER_NAME          STOCK_QUANTITY STOCK_STATUS
-----
```

8. Unavailable products

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     s.Seller_Name,
  5     i.Stock_Quantity,
  6     i.Stock_Status
  7 FROM AJIOInventory i
  8 JOIN AJIOProduct p
  9 ON i.Product_ID = p.Product_ID
 10 JOIN AJIOSeller s
 11 ON i.Seller_ID = s.Seller_ID
 12 WHERE i.Stock_Status = 'Unavailable';
```

PRODUCT_ID	PRODUCT_NAME	SELLER_NAME	STOCK_QUANTITY	STOCK_STATUS
103	Running Shoes	TRENDY WEAR	0	Unavailable

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
SELLER_NAME          STOCK_QUANTITY STOCK_STATUS
-----
103
Running Shoes
TRENDY WEAR          0 Unavailable
```

9. Available and Unavailable Product Count

```
SQL> SELECT
  2     Stock_Status,
  3     COUNT(*) AS Total_Products
  4 FROM AJIOInventory
  5 GROUP BY Stock_Status
  6 ORDER BY Stock_Status;
```

STOCK_STATUS	TOTAL_PRODUCTS
Available	2
Unavailable	1

10. Inventory Summary

```
SQL> SELECT
  2     COUNT(*) AS Total_Inventory_Records,
  3     SUM(Stock_Quantity) AS Total_Stock,
  4     SUM(
  5         CASE
  6             WHEN Stock_Status = 'Available' THEN 1
  7             ELSE 0
  8         END
  9     ) AS Available_Products,
 10     SUM(
 11         CASE
 12             WHEN Stock_Status = 'Unavailable' THEN 1
 13             ELSE 0
 14         END
 15     ) AS Unavailable_Products
 16 FROM AJIOInventory;
```

TOTAL_INVENTORY_RECORDS	TOTAL_STOCK	AVAILABLE_PRODUCTS	UNAVAILABLE_PRODUCTS
3	85	2	1